

Auto Insurance Claims Fraud Analysis

FROM: Tomorrow Lake, Data Analyst
RE: SIU Claims Routing Recommendations Based on Fraud Pattern Analysis
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Summary

I looked at 1,000 auto insurance claims to figure out which ones a Special Investigations Unit should review first. The dataset had a 24.7 percent fraud rate overall, but fraud is not spread evenly across claims. Some patterns are much stronger predictors than others, and combining a few of them gives you a simple risk score that an SIU could use to triage cases right away.

The headline finding is that Major Damage incidents have a 60.5 percent fraud rate, which is about six times the rate for other severity levels. Combining severity with three other red-flag indicators lets the SIU route 27.5 percent of claim volume and capture 38.1 percent of fraud, which works out to about \$5.5 million in flagged fraudulent claims.

Key Findings

- **Major Damage drives most of the fraud.** Major Damage claims have a 60.5 percent fraud rate, compared to 7 to 13 percent for Minor Damage, Total Loss, and Trivial Damage. This single attribute is the strongest individual signal in the dataset.
- **Red flags add up in a clean linear way.** I built a simple count of four red-flag indicators (severity, certain incident types, low witness counts, missing police reports). Each additional flag pushes the fraud rate higher in a predictable pattern, from 14.2 percent at 0 flags to 41.0 percent at 3 or 4 flags.
- **A 2-flag threshold gives the best trade-off.** Routing claims with 2 or more flags to SIU sends 27.5 percent of volume for review and captures 38.1 percent of fraud. Tighter thresholds send fewer claims but miss more fraud. Looser thresholds catch more fraud but overload investigators.
- **Single Vehicle and Multi-Vehicle Collision incidents have above-average fraud rates.** Incident type is a useful secondary signal that helps confirm a Major Damage flag.

Red-Flag Distribution

Red-Flag Count	Share of Claims	Fraud Rate
0 flags	47.3%	14.2%
1 flag	25.2%	25.2%
2 flags	16.8%	33.1%
3 or 4 flags	10.7%	41.0%

Recommendations

- **Route claims with 2 or more red flags to SIU automatically.** This is the rule that gives the best trade-off between investigator workload and fraud capture in the dataset I analyzed.
- **Treat Major Damage claims as a priority queue inside SIU.** Even within the 2-or-more-flags group, Major Damage claims have the highest expected fraud rate and should get attention first.

- **Track the false positive rate over time.** Any threshold creates some false positives. The team should monitor how often legitimate claims get held up so the threshold can be tuned if customer experience starts to suffer.
- **Revisit the threshold yearly.** Fraud patterns shift over time. The 2-flag rule is a good starting point but should be reviewed annually using the next year of claims data.

How I Did the Analysis

Source data: 1,000 auto insurance claims with 24.7 percent labeled as fraudulent. The dataset had attributes for incident severity, collision type, witness count, police report status, and several others.

Method: I profiled fraud rates across each attribute to identify the strongest individual signals, combined the top four into a simple flag count, and then calculated the trade-off curve between investigator workload and fraud capture for each possible routing threshold.

Tools: SQL for cleaning and exploration, Tableau for visualization and dashboard design.

Limitations: This analysis used 1,000 historical claims, which is a useful starting sample but not large enough to detect rare fraud patterns. The recommended routing rule should be validated against a larger and more recent dataset before being used in production. Like any rule-based screening tool, it will produce some false positives that need human review.

Supporting Materials

- Interactive dashboard:
public.tableau.com/app/profile/tomorrow.lake5414/viz/auto_insurance_dashboard/AutoInsuranceClaimsDashboar
- Code, data, and queries: github.com/tomorrowlake3/Auto_Insurance_Claims_Analysis